

Assignment 2: Jacobians

16-384: Robot Kinematics and Dynamics

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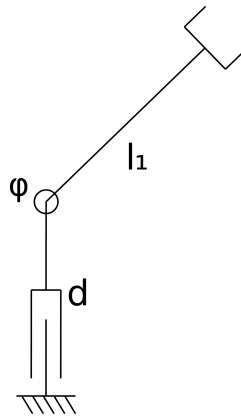
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1 Theory Questions

1) PR Arm

20 points

Use the PR Arm illustrated below from Assignment 1. The rotary joint φ measures the bearing angle, so $\varphi = 0$ means the arm is pointing straight up. The illustration shows the arm around $\varphi = \pi/4$ with $d = 1$ and $l_1 = 1$.



- a. [1 point] What is the dimension of the end-effector Jacobian J ? Assume the workspace is $SE(2)$.

- b. [1 points] What does each row of J mean (in 1–2 sentences)?

- c. [1 points] What does each column of J mean (in 1–2 sentences)?

- d. [2 points] Find the forward-kinematic map to the end effector for this arm for x , y , and θ as a function $f(d, \varphi)$. Check that the coordinates match the figure above for $d = 1$ and $\varphi = \pi/4$.

- e. [5 points] Determine the Jacobian of the arm by taking the partial derivatives of the forward-kinematic map with respect to d and φ .

- f. [5 points] Let $d = 0$, $\varphi = 0$ and $\dot{d} = \dot{\varphi} = v$, what is the contribution of each joint to the end-effector's linear velocity? To the angular velocity?

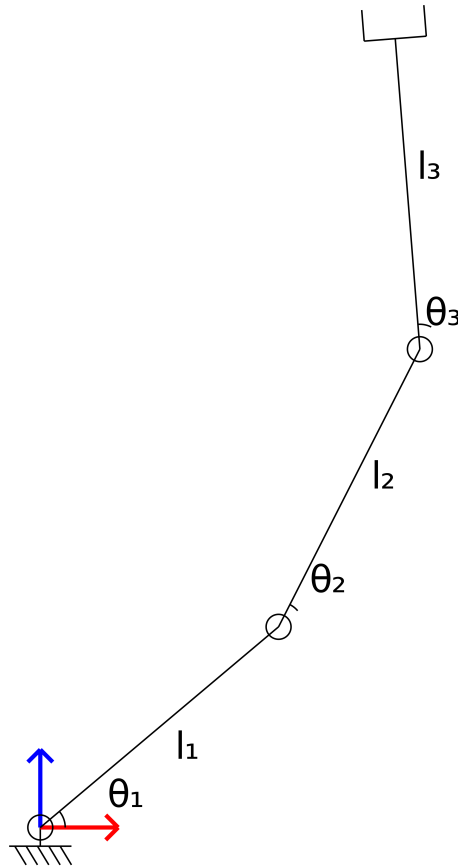
- g. [5 points] For this problem, assume that forces due to gravity are pointing in the negative y direction. Assume each **link** has a mass of 0.5 kg and the center of mass is in the center of that link.

- (i) What are the efforts (as a function of the configuration $\Theta = \begin{bmatrix} d \\ \varphi \end{bmatrix}$) necessary to counteract the effect of gravity on the first link? Assume all of the first link's mass is on the distal end of the prismatic joint.
- (ii) Of link l_1 ?
- (iii) What efforts are necessary to counteract the effect of gravity on all both links at the same time?

2) Robot Analysis: RRR

30 points

Consider the following robot. Assume no joint limits, and that the links lengths are $l_1 = l_2 = l_3 = 1$. The joint angles are measured as a relative angle from the previous link / horizontal, so $\theta_1 = \theta_2 = \theta_3 = 0$ means the arm is fully extended to the right. Note that positive angles rotate counter-clockwise, unlike problem 1.



- a. [8 points] Find the Jacobian for the end effector as a function of the joint angles θ_1 , θ_2 , and θ_3 . Assume the workspace is $SE(2)$.

- b. [2 point] Is this robot ever singular? If so, when?

- c. [5 points] When the arm is at position $(\theta_1, \theta_2, \theta_3) = (0, 0, \pi/2)$, and $\dot{\theta}_1 = \dot{\theta}_2 = \dot{\theta}_3 = v$, what is the contribution of each joint to the end-effector's linear velocity? To the angular velocity?

- d. [5 points] At configuration $\theta_1 = \theta_2 = \theta_3 = 0$,
- (i) Is this configuration singular? Why or why not?
 - (ii) What $\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$ can be applied to instantaneously move the end-effector in the planar direction $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$?
 - (iii) How about $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$?
 - (iv) With rotation, what joint velocities must be applied to move instantaneously in $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, where the third term is rotation?

- e. [5 points] At configuration $(\theta_1, \theta_2, \theta_3) = (0, \pi/2, 0)$, what joint torques are necessary to exert a wrench of $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$?

- f. [5 points] For this problem, assume that forces due to gravity are pointing in the negative y direction. Assume each **link** has a mass of 1 kg and the center of mass is in the center of that link.

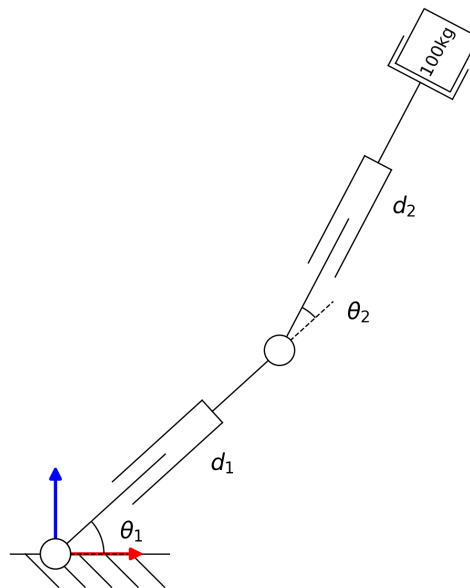
- (i) What are the joint torques (as a function of the configuration $\Theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$) necessary to counteract the effect of gravity on the center of mass of link l_1 ?
- (ii) Of link l_2 ?
- (iii) Of link l_3 ?
- (iv) What torques are necessary to counteract the effect of gravity on all three links at the same time?



3) Robot Analysis: RPRP Arm

25 points

Consider the following robot (assume no joint limits, and that the links do not intersect with the ground):



- a. [5 points] Find the forward kinematic map for the end effector.

- b. [5 points] Find the Jacobian for the end effector.

- c. [5 points] For this problem, assume that forces due to gravity are pointing in the negative y direction. Each link's mass is negligible (0 kg) and the end effector has a mass of

100 kg. Calculate the joint torques/efforts (as a function of the configuration $\Theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ d_1 \\ d_2 \end{bmatrix}$) necessary to counteract the effect of gravity.

- d. [5 points] At the configuration $(\theta_1, \theta_2, d_1, d_2) = (0, \pi/2, 1, 1)$, suppose we have joint torques/efforts of $(\tau_{\theta_1}, \tau_{\theta_2}, \tau_{d_1}, \tau_{d_2}) = (-1, -1, 1, 0)$ applied to the robot's joints. Solving the formula $\tau = J^\top f$ (if possible), what is the wrench at the end effector? What does this mean physically?

- e. [5 points] At the configuration $(\theta_1, \theta_2, d_1, d_2) = (0, \pi/2, 1, 1)$, suppose we have joint torques/efforts of $(\tau_{\theta_1}, \tau_{\theta_2}, \tau_{d_1}, \tau_{d_2}) = (1, 0, 0, 0)$ applied to the robot's joints. Solving the formula $\tau = J^\top F$ (if possible), what is the wrench at the end effector? What does this mean physically?

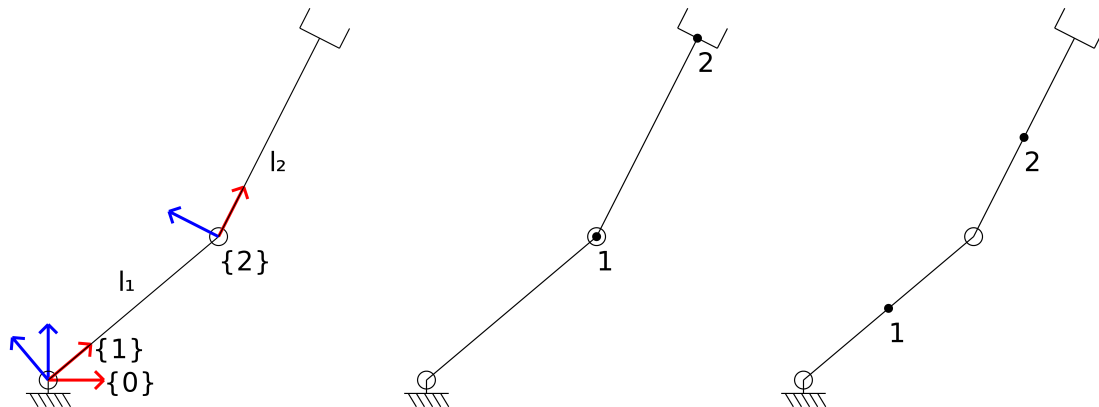
2 Code Questions

In these problems, we'll analytically test some of the work from the previous section. There is a provided `autograder.py` script which will check your code locally for correctness.

4) Jacobians and Velocities

25 points

Begin by opening the `ex1` directory in your Python environment. In this exercise, you must fill out several Python files. When completed correctly, the first three functions will define the forward kinematics and Jacobian to several points on an RR arm; the final function will provide a self-check for the first three functions.



- `forward_kinematics_RR.py`: As in Exercise 1 from Assignment 1, the forward kinematics to several frames should be filled in. Note: the frames for this problem are a subset of those from the previous assignment. In the code, H_{1_0} , H_{2_0} , and H_{3_0} refer to H_1^0 , H_2^0 , and H_3^0 respectively (where $\{0\}$, $\{1\}$, and $\{2\}$ are shown in the above left figure). $\{3\}$ is at the end-effector.
- `jacobian_link_ends_RR.py`: Jacobian matrices to the end of the robot links should be filled in in this problem. In the code, J_{END_1} and J_{END_2} refer to the Jacobians to the end of link 1 and 2 respectively (i.e., points 1 and 2 in the above middle figure).
- `jacobian_coms_RR.py`: Jacobian matrices to the centers of the robot links should be filled in in this problem. In the code, J_{COM_1} and J_{COM_2} refer to the Jacobians to the center of link 1 and 2 respectively (i.e., points 1 and 2 in the above right figure).
- `sample_path.py`: This function loads data taken from the robot, and draws the path taken by the center and end of each link. You should use the functions you have written above to fill in the position and velocity of these points.

After completing these tasks, run the `sample_path.py` function. You should see four separate lines, representing the path of the center and end of each link. These lines should be formed of arrows, which represent the velocity of each of these points. **Important:** if the arrows are not tangent to the path, there is an error in your code! Verify your code is correct before continuing. As before, when debugging, remember to use simple test cases such as

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ \frac{\pi}{2} \end{bmatrix} \text{ to check your matrices.}$$

The deliverables for this problem are the four Python files mentioned above, as well as the resulting figure from running `sample_path.py`. This figure should be saved as `path.png`. Additionally, `sample_path.py` will generate a `results.pkl` file containing the calculated data for each part of the robot. This file will be used for grading, so ensure it is generated correctly.

5) Forces and Gravity Compensation

25 points

Begin by opening the `ex2` directory in your Python environment. In this exercise, you must use the Jacobians found in the first problem to compute joint torques which result in forces applied by the robot. This will involve completing two functions:

- `get_joint_torques.py`: In this function, you must determine the joint torques necessary for applying a given force with the end effector (e.g., centered at the end effector origin in a given direction with a given magnitude).
- `get_grav_comp_torques.py`: In this function, you will compute the joint torques which compensate for the force of gravity on each part of the robot, given the current joint angles. Note the direction and magnitude of acceleration due to gravity is passed into this function (assume units of m/s^2).

After completing these functions, run the `sample_torques.py` script. This script will generate plots of your computed torques for a test run of the robot. Two figures will be produced: one containing torques for applying a force with the end effector, and the other containing torques for compensating for gravity.

The deliverables for this problem are the two completed Python files above, as well as the resulting figures from running `sample_torques.py`. These figures will be automatically saved as `torques_ee.png` and `torques_grav_comp.png`, respectively. Please also include copies of these figures in your writeup!

Additionally, the `sample_torques.py` script will save your computed torques as pickle files (`tau_desired_force.pkl` and `tau_grav_comp.pkl`). These will be used for grading purposes, so ensure they are generated correctly.

Note: The `sample_torques.py` script uses functions from the `ex1` directory. Make sure your Python environment is set up correctly to access these functions.

6) Submission

To submit, run `create_submission.py`. It will first run `autograder.py` to confirm your submitted files run without error and then produce a zip file `submission.zip` containing your `ex1` and `ex2` folders. Rename that file to `<andrew_id>.zip` and upload it to Gradescope.

3 Submission Checklist

- ☐ Create a PDF of your answers to the written questions with the name `<andrew_id>.pdf` and submit it to Gradescope.
- ☐ Run `autograder.py` and confirm it reports no failures.
- ☐ Run `create_submission.py` in the python terminal.
- ☐ Rename `submission.zip` and upload `<andrew_id>.zip` to Gradescope.
- ☐ Coding questions are autograded, with no limit on number of submissions within the mandated deadline.